

Question	Marking Guidance	Mark	Comments
07.1	Sucrose;	1 (1 x AO1)	

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07.2	1. Insoluble (in water), so doesn't affect water potential/osmosis; 2. Coiled/spiral/(α -)helix, so compact OR Coiled/spiral/(α -)helix so can fit many (molecules) in small area; 3. Many (α -)glucose for respiration; 4. Branched/more ends so fast breakdown/hydrolysis; 5. Large (molecule), so can't leave the cell;	3 max (3 x AO1)	1. Accept Ψ for water potential 4. Ignore surface area 4. Accept branched so glucose readily released
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07.3	1. Some sugars converted to cellulose/starch; 2. Some sugars used in respiration; 3. Some sugars not moved down; 4. Some sugars moved further down than E;	1 max (1 x AO2)	1, 2, 3 and 4. Accept 'organic substance' for sugars or any suitably named organic substance for sugars, e.g. glucose 3. Accept some sugars remain in the leaf
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07.4	(In support of mass flow hypothesis) 1. Radioactivity is higher in phloem than xylem; 2. Radioactivity found in sections below leaf so it moved down stem/plant; 3. Leaf is source OR Leaf produces organic substances; (Against mass flow hypothesis – max 3) 4. Some radioactivity present in xylem; 5. Only tested in one species OR Only tested in geraniums; 6. No evidence for transport above the leaf; 7. Movement could be due to gravity;	4 max (4 x AO3)	3. Accept leaf produces any named organic substance 6. Accept other leaves/fruit/flowers